

Course Code: INT446

GENERATIVE AI WITH LARGE LANGUAGE MODELS

Course objectives:

The course helps the learners to Analyze the basic concepts of Large Language Models and its applications in Natural Language Processing and apply the Large Language Models in various *language-related applications* such as text classification, language translation, content generation, ChatBot, etc.

UNIT - I 15 Periods Generative AI Use Cases, Fundamentals, and Project Life Cycle:

Use Cases and Tasks - Foundation Models and Model Hubs -Generative AI Project Life Cycle -Generative AI on AWS -Why Generative AI on AWS? - Building Generative AI Applications on AWS

Prompt Engineering and In-Context Learning: Prompts and Completions-Tokens -Prompt Engineering -Prompt Structure -In-Context Learning with Few-Shot Inference -Prompt Engineering Best Practices -Inference Configuration Parameters

Large-Language Foundation Models: Large-Language Foundation Models -Tokenizers - Embedding Vectors -Transformer Architecture -Types of Transformer-Based Foundation Models -Pretraining Datasets -Scaling Laws -Compute-Optimal Models

UNIT - II 15 Periods Memory and Compute Optimizations: Memory Challenges -Data Types and Numerical Precision- Quantization Optimizing the Self-Attention Layers - Distributed Computing - Distributed Computing on AWS

Fine-Tuning and Evaluation: Instruction Fine-Tuning - Instruction Dataset - Instruction Fine Tuning – Evaluation

UNIT - III 15 Periods Parameter-Efficient Fine-Tuning: Full Fine-Tuning Versus PEFT - LoRA and QLoRA Prompt Tuning and Soft Prompts

Fine-Tuning with Reinforcement Learning from Human Feedback: Human Alignment: Helpful, Honest, and Harmless - Reinforcement Learning Overview - Train a Custom Reward Model- Existing Reward Model: Toxicity Detector by Meta- Fine-Tune with Reinforcement Learning from Human Feedback - Evaluate RLHF Fine-Tuned Model

UNIT - IV 15 Periods Model Deployment Optimizations: Model Optimizations for Inference - Large Model Inference Container - AWS Inferentia: Purpose-Built Hardware for Inference - Model Update and Deployment Strategies - Metrics and Monitoring- Auto scaling

Context-Aware Reasoning Applications Using RAG and Agents: Large Language Model Limitations - Retrieval-Augmented Generation –RAG Orchestration and Implementation Agents - Generative AI Applications -FMOps: Operationalizing the Generative AI Project Life Cycle

Multimodal Foundation Models: Use Cases-Multimodal Prompt Engineering Best Practices Image Generation and Enhancement - Inpainting, Outpainting, Depth-to-Image- Image Captioning and Visual Question Answering- Model Evaluation- Diffusion Architecture Fundamentals- Stable Diffusion 2 Architecture - Stable Diffusion XL Architecture.

REFERENCES

1. Chris Fregly, Antje Barth & Shelbee Eigenbrode, Generative AI on AWS, 1st Edition, O'Reilly Media, 2023
2. Ben Auffarth, Generative AI with lang chain, 1st Edition, Packt Publishing, 2023.
3. Gerhard Paaß and Sven Giesselbach, Foundation Models for Natural Language Processing, Springer International Publishing, 2023

ONLINE MATERIAL

<http://www.digimat.in/nptel/courses/video/106106211/L01.html>

LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to:

- UNIT - I □ Analyze the basics of Foundation models and prompting techniques
- UNIT - II □ Design the fine-tune Large language models using PEFT and RLHF
- UNIT - III □ Apply the quantization and distributed training to minimize the required GPU, RAM and scale model training horizontally across multiple GPUs for larger models.
- UNIT- IV □ Design and analyze the frameworks such as LangChain, ReAct, and PAL